

UTKARSH GUPTA

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EDUCATION

University of Southern California <i>Master of Science in Computer Science</i> <i>Relevant Coursework: DL for Robotic Manipulation, Robotic Perception, Autonomous Systems, Robotic Learning</i>	Aug 2024 – May 2026 (GPA: 3.9/4.0)
Maharashtra Institute of Technology, Pune <i>Bachelor of Technology in Computer Science and Engineering</i>	Dec 2020 - Aug 2024 (GPA: 3.8/4.0)

SKILLS

Languages and Platforms: Python, C++, ROS2, CUDA (C++), Linux
Robotic Perception: 3D Data Processing, Sensor Fusion (Camera + LiDAR), 3D Reconstruction, Digital Twins
Planning, Localization and Simulation: Nav2, RRT, PRM, A*, Pose Estimation, EKF, SLAM, Mujoco, NVIDIA Omniverse
Machine Learning: Vision-Language-Action (RT-X, Octo), Segmentation, Classification, Reinforcement Learning
Libraries and Framework: PyTorch, TensorFlow, Keras, OpenCV, SciPy, NumPy, scikit-learn

RELEVANT EXPERIENCE

Autonomy Engineering Intern <i>Advanced Drone Solutions</i> - Integration of custom-programmed solutions within ADS's autonomy platform for their drone platform - R&D of autonomy-related packages, frameworks, and ML components for improved vision pipeline performance	Jan 2026 - Present
Research Assistant <i>PSI Lab, USC</i> - Developing simulated data collection pipelines in IsaacSim and Mujoco for real2sim robotics tasks for VLA training (link) - Made sim deployment scripts for different embodiments for data variety with domain randomization and force feedback - Integrating both heuristic and VLA models for robot decision making for trajectory collection and parallel randomization	Oct 2025 - Present
Research Assistant <i>CPS Vida Lab, USC</i> - Working on interpretability techniques for VLA models to improve task failure analysis and task alignment for safety - Used Human-in-the-loop policies for VLAs to boost action success rate by 20%, enhancing HRI for low confidence states - Developed imitation learning strategies for MAPF, accelerating policy convergence by 30% for collaborative tasks	Nov 2024 - Nov 2025
Robotics Engineering Intern <i>ERIC Robotics</i> - Engineered real-time railtrack and 3D safety algorithms, improving AGV navigation safety by 25% for dynamic environments Cut AMR initial pose estimation and localization time by 40% by optimizing pose estimation with LiDAR scans - Developed and deployed perception and navigation algorithms on Jetson AGX Orin, optimizing CUDA-accelerated libraries for real-time performance	May 2025 - Aug 2025

PROJECTS

Digital Twin Pipeline via 3D Reconstruction Digital Twin, 3D Reconstruction, Photogrammetry - Architected a 3D reconstruction pipeline to generate high-fidelity models of industrial components, - Generated reconstruction for partial (180°-250°) image scans by applying Poisson surface reconstruction
Camera-Only ADAS for Collision Avoidance and Rerouting OpenCV, PyTorch, YOLOv5, DeepLabV3, StereoSGBM, KITTI Dataset - Developed a camera-only ADAS with 85% obstacle detection accuracy up to 50 meters for KITTI stereo dataset - Fused depth data with YOLOv5 for 3D localization, enabling real-time rerouting decisions in <100ms. - Implemented A* pathfinding on a dynamic grid, preventing 81% of simulated collisions and ensuring safety
RL-Based Stealth Navigation in Multi-Agent Environments Python, Gym, PyTorch, Matplotlib, Stable-Baselines3 Engineered a multi-agent system and gym environment where agents achieved an 85% goal success rate in stealth tasks.